The winch model in AWETrim

Force law, drum inertia and drivetrain friction

1 Model

The kite is quasi-steady: at each instant its force balance gives the tether tension T at the ground for a given reel speed v_r. The winch closes the radial direction. Between the tether and the motor sits the drivetrain: drum, gearbox and motor, all referred to the tether. The drum obeys

$$m\,\dot{v}_r = T - F_f(v_r, T) - F_m, \tag{1}$$

with F_m the motor (generator) force, m the rotating inertia seen at the tether and F_f the friction:

$$m = J\left(\frac{n}{r_d}\right)^2, \qquad F_f = (F_c + k\,T)\tanh\!\left(\frac{v_r}{\varepsilon}\right) + c_v\,v_r. \tag{2}$$

J is the total inertia on the motor side, n the gear ratio and r_d the drum radius. F_c is the Coulomb friction, c_v the viscous coefficient and k an optional load-proportional part (pulleys, belt, gear mesh). The tanh, with $\varepsilon = 0.05\,\mathrm{m\,s^{-1}}$, smooths the sign of v_r, so the model stays twice differentiable where v_r changes sign.

The force law sets the motor force. The winch controller is a torque-controlled machine: its law $F_{\mathrm{law}}(v_r, u_p)$ (linear or quadratic in v_r, soft-capped at the force limits, offset by the depower u_p) is the force the motor applies. With Eq. (1), the tension the kite feels is

$$T = F_{\mathrm{law}}(v_r) + F_f(v_r) + m\,\dot{v}_r. \tag{3}$$

Friction raises the tension during reel-out and lowers it during reel-in; inertia adds tension while the drum accelerates. With $m = F_c = c_v = k = 0$ (the default) Eq. (3) is the ideal winch, $T = F_{\mathrm{law}}$.

Shaft power. The power at the motor is

$$P_m = F_m v_r = T v_r - \underbrace{v_r F_f}_{\geq 0} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\tfrac{1}{2} m v_r^2\right). \tag{4}$$

Friction is lost in both reel directions. The inertia term is the drum's kinetic energy: it is returned over a closed cycle, so on a periodic cycle inertia changes the mean power only through the force limits.

2 Discretisation

Forward simulation. The march solves, at each node, the kite's force balance and Eq. (3) for $(T, u_s, \dot{s}, v_r)$. The drum's time constant, m over the slope of the law ($\sim$0.06 s for the LEI-V3), is far shorter than a node step ($\sim$0.3 s), so $\dot{v}_r$ is an implicit difference over the step that ends at the node, solved with it; the first node starts with a steady drum. Two schemes (`time_scheme`): `euler`, the backward difference $\dot{v}_r \approx (v_{r,i} - v_{r,i-1})/\Delta t$ with the step $\Delta t = \Delta s/\dot{s}_{i-1}$ (first order), and `bdf2`, the variable-step BDF2 difference over the last two steps with the trapezoidal step $h = 2\Delta s/(\dot{s}_{i-1} + \dot{s}_i)$ and tether length $r_i = r_{i-1} + h\,(v_{r,i-1} + v_{r,i})/2$ (second order; like backward Euler it damps the stiff drum mode). The default is `euler` quasi-steady and `bdf2` when the kite is dynamic too.

Optimisation. The cycle NLP uses the same model and scheme. Its objective is the mean shaft power, with the inertia part written as the exact change of $\frac{1}{2}mv_r^2$ over each interval, so it telescopes. With the force law (`winch_mode: force_law`) Eq. (3) replaces the law row. With a free reel speed (`free_speed`) the motor force $F_m = T - F_f - m\dot{v}_r$ is held below `max_tether_force` whenever the drivetrain is not ideal (friction adds to it on reel-in).

Resolution. Table 1 marches one LEI-V3 figure eight at $9\,\mathrm{m\,s^{-1}}$ (tether inertia and winch friction on) at 25 to 400 nodes, against the finest `bdf2` run of the same model. The quasi-steady march is accurate from 25 nodes in either scheme; `bdf2` halves its small power error. The dynamic march converges at about first order with `euler` and overstates the power at coarse steps (the scheme's error, not physics), but at close to second order with `bdf2`, which at 100 nodes is as accurate in tension as `euler` at 400. Converged, the dynamic model's mean power is 0.15 % below the quasi-steady one; its tension peaks are ∼0.3 kN higher (drum and kite inertia), at every resolution. A whole pumping cycle (same wind and model) converges more slowly, the dynamics acting at the reel-out and reel-in transitions, where the depower profile and the force law have corners: the dynamic shaft power is +3.2, +1.7, +1.1 and +0.9 % above the quasi-steady one at 114, 228, 456 and 912 nodes, and its tension peak settles at 8.93 kN from 456 nodes (quasi-steady 8.62 kN at every resolution). A dynamic cycle therefore needs a step of about 0.2 s, two to four times finer than the eight. A node costs ∼10 ms quasi-steady and ∼14 ms dynamic; `bdf2` adds 5–10 %.

Table 1: Mean shaft power error and largest tension difference against the 400-node `bdf2` run of the same model, and the observed order of the tension difference between successive halvings.

Model	Scheme	Nodes	ΔP (%)	max $\lvert\Delta T\rvert$ (N)	Order
quasi-steady	`euler`	25	−0.41	2.6	
		100	−0.12	0.29	1.3
		400	−0.02	0.005	
	`bdf2`	25	−0.16	2.6	
		100	−0.05	0.28	1.3
dynamic	`euler`	25	+2.92	333	
		100	+0.73	148	0.7
		400	+0.20	43	0.9
	`bdf2`	25	−0.34	323	
		100	−0.09	44	1.6
		200	−0.02	13	1.7

3 Tether inertia

The lumped tether (straight, inextensible, uniform, mass $m_t = \mu r$, pinned at the anchor) also carries the inertia of its own mass. Its material moves along it at v_r and turns with $\boldsymbol{\Omega} = \mathbf{e}_r \times \dot{\mathbf{e}}_r$, $|\boldsymbol{\Omega}| = v_\tau/r$ (spin about the tether moves no material). A point at distance ρ accelerates as

$$\mathbf{a}(\rho) = (\dot{v}_r - \rho\, v_\tau^2/r^2)\,\mathbf{e}_z + \big[2v_r v_\tau/r + \rho\,\tfrac{\mathrm{d}}{\mathrm{d}t}(v_\tau/r)\big]\mathbf{e}_x + \rho\,(v_\tau/r)\,\omega_z\,\mathbf{e}_y, \tag{5}$$

in the course frame (tangential x, normal y, radial z), ω_z the frame's turn rate about the tether. Integrating along the tether, with moments about the anchor for the cross components (tether reeled at the anchor carries no angular momentum), the load it takes from the kite is

$$\mathbf{F}_t = -\Big[\tfrac{m_t}{3}\big(\dot{v}_\tau + \tfrac{2v_\tau v_r}{r}\big),\ \tfrac{m_t}{3}\,v_\tau\,\omega_z,\ m_t\big(\dot{v}_r - \tfrac{v_\tau^2}{2r}\big)\Big]^{\mathsf{T}}. \tag{6}$$

Across itself the tether acts as $m_t/3$ at the kite (with twice the kite's Coriolis-type term, from reeling), along itself as m_t (with half its centrifugal term). Quasi-steady only the centrifugal,

Coriolis and turn terms remain. It is on by default (`inertia: false` in the tether block drops it), in the ROM path and in the aerostructural trim alike.

4 Parameters and use

Table 2: LEI-V3 drivetrain: the TU Delft GS1 drivetrain as V3Kite.jl flies it (`sim_settings_reelout.yaml`, the KiteUtils defaults; Fechner 2016, Table 4.1). Stored tether-referred in `system.yaml`, `drums[0]`.

Quantity	`system.yaml` key	Value
Inertia at the tether, $0.204\,(6.2/0.1615)^2$	`inertia_equivalent_mass`	300.7 kg
Coulomb friction F_c	`friction_coulomb`	122 N
Viscous friction c_v	`friction_viscous`	30.6 N s m^{-1}
Load fraction k (not in V3Kite)	`friction_load_fraction`	0.0535

The drivetrain is off unless `sim_parameters` asks for it: `winch_drivetrain: true` takes m, F_c and c_v from the system file; `winch_inertia_mass`, `winch_friction` and `drivetrain_load_fraction` set the parts one by one (`true` or a value). The cycle script takes `--winch-drivetrain`. Code: `awetrim.system.winch.Drivetrain` and `Winch.radial_equation`.

5 Example

Figure 1 marches the 9 m s^{-1} LEI-V3 cycle with its force law, without optimising. The motor force stays at the law's limit while the tension reaches 8.87 kN, the extra being friction (∼0.23 kN in reel-out) and the drum accelerating into reel-out (up to 0.74 kN). The mean shaft power is 9.8 kW, against 11.0 kW at the tether.

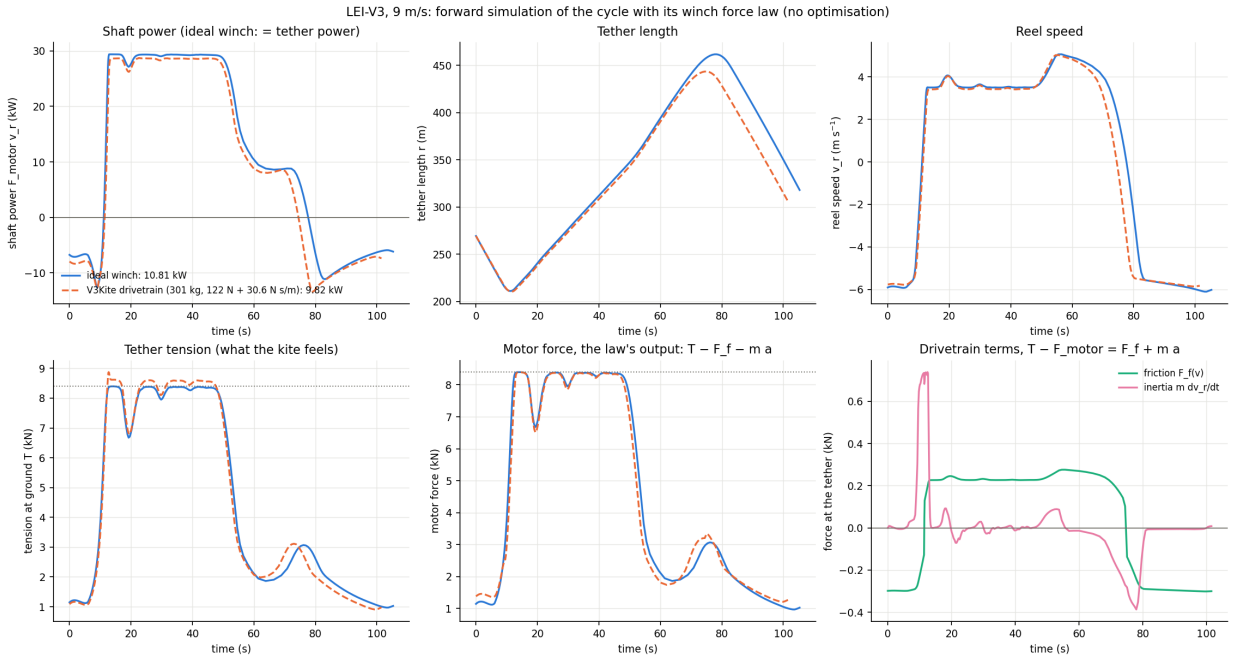


Figure 1: Forward simulation of the 9 m s^{-1} cycle path with an ideal winch and with the LEI-V3 drivetrain (Table 1).